

Appendix: Inventory of Existing IVB Simulation Designs

Task 10 static audit

2026-07-11

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1 Scope and reproducibility

This appendix is a static audit of simulations already present in the repository. It does not run, source, or modify a Monte Carlo script. The companion builder, `build_simulation_design_inventory.R`, reads only source files and existing artifacts, verifies declared paths, records MD5 hashes, and writes the inventory and manifest. Equations and parameter statements labelled as verified are transcribed from the versioned scripts; parameter grids retained in timing/grid CSVs are referenced rather than re-created. Any unlinked legacy script or runtime-configurable repetition count is explicitly marked as a gap.

The inventory uses a deliberately narrow evidentiary rule: a result is *directly verified* only when its source script and an existing result artifact both exist. It does not certify that a CSV was regenerated in this audit, nor does it infer a causal interpretation from a coefficient shift alone.

2 Coverage matrix

Table 1: Table 1. Required coverage and the directly mapped inventory rows.

required_coverage	inventory_rows
dual_role	diag_heterogeneous; diag_ivb_sign; diag_sensitivity; diag_shortT; diag_timeFE; dual_role_8models; dual_role_asymmetry; dual_role_base; dual_role_firewall; dual_role_varyT; dual_role_varyT_8models; mechC_adl
firewall	callaway_controls; diag_sensitivity; diag_shortT; direct_carryover; direct_feedback; dual_role_8models; dual_role_firewall; dual_role_varyT; dual_role_varyT_8models; feedback_Y_to_D; feedback_carryover; mechC_adl; nl_carryover; nl_collider; nl_interact; persistent_unobserved; staggered_posttreat
mediator_overcontrol	intsel_twoperiod; overcontrol_contemp; overcontrol_full
nonlinearity	callaway_controls; nl_carryover; nl_collider; nl_interact; overcontrol_contemp; overcontrol_full
feedback	diag_phi; direct_feedback; feedback_Y_to_D; feedback_carryover
carryover	direct_carryover; feedback_carryover; nl_carryover
staggered_binary_treatment	staggered_posttreat; v4_mechC

required_coverage	inventory_rows
measurement_error	mechD
callaway_style_controls	callaway_controls
contemporaneous_jensen_robust_periodic_persistent_unobserved	

3 Design inventory

Table 2: Table 2. Existing simulation DGPs/families and result artifacts.

inventory_id	family	causal	design	description	N	Value	repetitions	process	effect	estimator	code	script	file	path	status	notes
callaway	dyn	dyn	Functional	delta=156	time	500	iid	unit/	shad	bias, simulations	dyns/d	dyns/d	dyns/d	dyns/d	dyns/d	dyns/d
style	in-	form	un-	sce-	CSV	Gaus-	FE	TWFE	BY	MDL/MCSE,						source
non-	cludes	or	der	nar-		sian	(which	model	dis-							com-
lin-	gamma	nlevel-	stated	ios x		in-	do	bat-	carded							ment
ear	g(Z_lag);	depend	DGP;	500		no-	not	tery.	draws,							distin-
con-	B: Y	trend	di-	reps;		va-	ab-		tim-							guishes
trol	in-	mis-	ag-	NL		tions;	sorb		ing.							this
re-	cludes	spec-	nos-	types		H_i~N(0,1).										from a
la-	lambda	ifica-	tic	soft-		x										col-
tion-	H_i(t/T),	tion;	of	poly2,	power1.5,	quadratic;	time).									lider
ship	Z in-	col-	spec-	ex-												mech-
and	cludes	lider	ifi-	act												anism.
level-	delta_H	on/off	ca-	grid												
depend	H_i;	via	tion	in												
trends	D/Z	delta_Y	ro-	tim-												
	remain		bust-	ing												
	dy-		ness.	CSV.												
	namic															
	dual-															
	role.															
diag	het	dyn	Dual-	mean	reps=200	reps=200	iid	unit/	FE	bias	simulations	dyns/d	dyns/d	dyns/d	dyns/d	dyns/d
het-	dual-	role	beta	sigma_beta={0,.3,.5,1}	Gaus-	FE	ADL, sum-									arti-
ero-	role	dy-	/	base/extended	sian.	in	ex-	maries.								fact,
ge-	base	nam-	esti-	sce-												not a
neous	DGP	ics	ma-	nario.												main
treat-	with	with	tor													planned
ment	beta_i	het-	tar-													DGP.
ef-	hetero-	ero-	get													
fects	geneity	ge-	un-													
	con-	neous	der													
	trolled	causal	het-													
	by	ef-	ero-													
	sigma_beta	fact.	gene-													
			ity.													

investor	type	family	causal	estimator	description	Nr. Values	repetition	process	effects	models	code	description	file_path	status	notes
diag_1pho	Dynamic	Lagged	beta=1	reps=200	uscausca	200	iid	unit/FE	bias	ADL	simulation	stons/diags/diags/	diag_1pho	flagged	artificial
IVB	dual-	con-	and	gamma_D={0,.05,.15,.3}	Gaus-	FE	short/long.								arti-
sign	role	found-	sign	other	sian.			sign							fact.
	base	ing/current	fit	grid				ta-							
	DGP	col-	spec-	in				bles.							
	vary-	lider.	ifi-	source.											
	ing		ca-												
	gamma_D		tion												
	and		shift.												
	delta_Y.														
diag_1pho	Dynamic	Lagged	beta=1	reps=300	uscausca	300	iid	unit/FE	bias	ADL	simulation	stons/diags/diags/	diag_1pho	flagged	artificial
feed-	system	feed-	con-	phi={0,.05,.10,.15};	Gaus-	FE	ADL, and								arti-
back	vary-	back	di-	rho_Y={0,.5}.	sian.	for	no- R2								fact.
mech-	ing	in	tional				FE FE di-								
a-	D_t	dual-	on				modeADL, ag-								
nism	re-	role	his-				D~Y lag-								
	sponse	dy-	tory.				aux- tics.								
	phi	nam-					il-								
	Y_lag.	ics.					iary.								
diag_1sens	Dynamic	Lagged	beta=1	reps=200	uscausca	200	iid	unit/FE	bias	ADL	simulation	stons/diags/diags/	diag_1sens	flagged	artificial
pa-	dual-	con-		rho_Y={.3,.5,.7};	Gaus-	FE	ADL and								
ram-	role	found-	sta-		sian.		vari- sta-								unsta-
e-	DGP	ing/current	ble				ants. bil-								ble are
ter	vary-	col-	sub-				ity								ex-
sen-	ing	lider.	set				sum-								cluded
si-	rho_Y,gamma_D,gamma_Y.						maries.								by
tiv-	and														source
ity	stabil-														logic;
	ity.														see
															source/result.
diag_1short	Dynamic	Lagged	beta=1	reps=200	uscausca	200	iid	unit/FE	bias	ADL	simulation	stons/diags/diags/	diag_1short	flagged	artificial
short-	dual-	con-	finite-	T={10,30};	Gaus-	FE	long; and								arti-
T /	role	found-	T	sce-	sian.	where	FE Nickell-								fact.
Nick-	DGP	ing/current	esti-	nar-			spec- ADL; cost								
ell	com-	col-	ma-	ios			i- no- com-								
	paring	lider.	tor	in			fied. FE par-								
	T=10		be-	source.			ADL. i-								
	and		hav-				son.								
	30.		ior.												
diag_1time	Dynamic	Dual-	beta=1	reps=200	uscausca	200	iid	unit/FE	bias	ADL	simulation	stons/diags/diags/	diag_1time	flagged	artificial
time	dual-	role		sigma_delta={0,.3,.5,.1}	Gaus-	FE	and sum-								arti-
FE	role	dy-	sce-		sian.		FE- maries.								fact.
in	DGP	nam-	nar-			time	ADL								
DGP	with	ics	ios			shocks	vari-								
	time-	plus	in			and	ants.								
	shock	com-	source.			in-									
	vari-	mon				no-									
	ance	time				va-									
	sigma_delta	shocks.				tions.									

investitor	simulation	family	causal	estimator	description	Nr. Variables	repetition	process	effect	models	descript	file	paths	notes
direct	linear	Dynamic	Lagged	contemporary	Normal	REPS=500	sc=500	iid	unit/Gaus-	sme bias, simulations	RMSE, MCSE,			call every coefficient bias to beta when the omitted carry-over changes the estimand.
di-rect	dual-role	Z-con-	beta=1	beta2	CSV				FE.	and MCSE,				feed-back is not con-
car-ry-	DGP	founder	current	rho_Z					sian.	ADL tim-				tempo-
over	with	col-	may	grid						ing.				ra-neous
	Y_t	lider	tar-	in										feed-
	includ-	plus	get	tim-										back.
	ing	di-	mis-	ing										
	beta	rect	spec-	CSV.										
	D_t+beta	2	ified											
	D_lag	ment	ef-											
	plus	car-	fects											
	lagged	ry-	when											
	Y/Z.	over.	omit-											
			ting											
			D_lag.											
direct	lagged	Dynamic	Y_{t-1}	beta=1	IN_REPS=500	sc=500	iid	unit/Gaus-	sme bias, simulations	RMSE, MCSE,				feed-
out-	role	>D_t	der	and	CSV				FE.	and MCSE,				back is
to-	DGP	feed-	stated	rho_Z					sian.	ADL tim-				not
treat-	with	back;	his-	grid						ants.				con-
feed-	D_t	lagged	tory	in										tem-
back	includ-	Z	ad-	tim-										pora-
	ing phi	con-	just-	ing										neous
	Y_lag.	found-	ment.	CSV.										feed-
		ing/current												back.
		col-												
		lider.												
dual	role	D+gamma	rho_Z	sigma_alpha	rho_Z={.1,.3,.5,.7,.85}	sigma_alphaZ={.5,2}	sc=500	iid	unit/Gaus-	sme bias, simulations	RMSE, MCSE,			and all
role	Z_lag+rho	founder							FE	short/long;				stated
Z	D_lag+u;	founder							sian	in TWFE				values
full	Y=alpha	Y+beta							in	FE				are
2x2x2	D+gamma	Z_Y							no-	es-				source
es-	Z_lag+rho	Y							va-	ti-				com-
ti-	Y_lag+e;	lider;							tions/	mit				ments/parameters.
ma-	Z=alpha	Z+delta_D							ef-	tors.				
tor	D+delta_Y	asts							fects.	with-				
grid	Y+rho_Z	FE/no-								out				
	Z_lag+nu	FE,								FE.				
		lag												
		in-												
		clu-												
		sion/exclusion.												

investigator	study	family	causal diagram	estimator	description	N	T	beta	repetition	fixed effects	instruments	models	description	file	paths	notes
dual	role	dy-	con-	beta=1	N/T/bu	source	source	500	iid	unit/TWFE	Bias	simulations	models	role	if/line	strengths/artifact
	causal	dynamic	founder;	and	grid				Gaus-	FE. and	and					source
	channel	dual-	Z_t	di-					sian.	ADL net						for
	asym-	role	col-	rectly						short/TWFE	ADL					exact
	me-	equa-	lider.	in							con-					grid;
	try	tions;		source;							trasts.					no raw
		varies		reps=500.												arti-
		D->Z														fact
		and														listed.
		Y->Z														
		chan-														
		nel														
		strengths.														
dual	role	base-	Z_lag	beta=1	N=100,	10030,	10020	100	iid	unit/TWFE	Bias	simulations	models	role	if/line	strengths/artifact
	role	Z_lag+rho	D	rho_Z					Gaus-	and short/long;						veri-
	Z	D_lag+u;	con-	and					sian	time ADL MCSE						fied
	base-	Y=alpha	founder	sigma_alpha	Z				in-	FE short/long.						source
	line	D+gamma	Z_Y	grid					no-	where	model.					and
		Z_lag+rho	Y	in					va-	ap-						CSV.
		Y_lag+e;	col-	re-					tions	pli-						
		Z=alpha	Z_delta_D	sult					and	ca-						
		D+delta	D_t-	CSV.					Gaus-	ble.						
		Y+rho_Z	Z_t<-						sian							
		Z_lag+nu	Y_t.						unit							
									ef-							
									fects.							
dual	role	fire-	Dual-	beta=1	Source	source	source	500	iid	unit/TWFE	Bias	simulations	models	role	if/line	strengths/artifact
	role	dy-	role		grid;				Gaus-	FE. and	and					N/T
	fire-	dynamic	Z		reps=500;				sian.	ADL fire-						values
	wall	dual-	with		rho_Y					short/long.						are
	sen-	role	fire-		in-						com-					source-
	si-	equa-	wall		cludes						par-					verified
	tiv-	tions;	through		0,.5,.8.						isons.					but re-
	ity	varies	ob-													ported
		out-	served													as
		come	his-													source
		persis-	tory.													to
		tence														avoid
		to test														duplicating
		whether														a pa-
		Y_lag														parameter
		blocks														table.
		inher-														
		ited														
		paths.														

investitor	equation	family	usual	estimator	description	Nr	Value	repetition	spread	effect	test	nodes	script	file	paths	notes		
dual	role	Static	Dual	beta=1	Source source	criss	200	iid	unit/	TWFE	Bias	simulations/	dual_role	black	file	output	leissin	
	finite-T	dynamic	role		grid			Gaus-	FE.	and	by					script		
	vari-	role	with		varies			sian.	ADL	T.	short/long.					de-		
	a-	equa-	lagged		T											clares		
	tion	tions;	con-		and											an		
		varies	found-		rho_Z;											output		
		T.	ing/current		reps=200.											path,		
			col-													but no		
			lider.													match-		
																ing		
																ver-		
																sioned		
																result		
																arti-		
																fact		
																exists		
																in the		
																au-		
																dited		
																tree; it		
																cannot		
																sup-		
																port a		
																nu-		
																meri-		
																cal		
																claim.		
dual	role	Static	Dual	beta=1	Source source	criss	500	iid	unit/	Pooled	TWFE	Bias	simulations/	dual_role	black	file	output	leissin
	finite-T	dynamic	role		grid			Gaus-	FE	2x2x2	SD,					only		
	full	role	with		varies			sian.	in	grid.	MCSE					arti-		
	es-	equa-	lagged		T			FE	by							fact;		
	ti-	tions;	con-		and			mod-	T.							raw		
	ma-	varies	found-		rho_Z;			els.								results		
	tor	T	ing/current		reps=500.											ab-		
	grid	under	lider.													sent.		
		eight																
		esti-																
		ma-																
		tors.																
feedback	K	to_D	Y_lag-	beta=1	N_REPS=500	criss	500	iid	unit/	time	bias,	simulations/	dual_role	black	file	output	leissin	
	Y_lagin-		>D_t	con-	stability	CSV		Gaus-	FE.	model	RMSE/MCSE,					veri-		
	to	cludes	and	di-	screened			sian;		TWFE/ADL						fied		
	D	phi	D_t-	tional	grid			un-		grid	bil-					source/results.		
	with	Y_lag;	>Z_t<-on	in	tim-			sta-		plus	ity/timing.							
	ex-	Y and	Y_t;	record	ing			able		aux-								
	tended	Z	Z_lag	his-	ing			pa-		il-								
	ADL	follow	af-	tory.	CSV.			ram-		iary.								
		dy-	fects					e-										
		namic	D,Y.					ter										
		dual-						sets										
		role						flagged.										
		equa-																
		tions.																

investitor	tid	family	causal	estimator	description	Nr	Value	repetition	seed	effect	test	models	descript	file	paths	notes
feed-back-and-car-ry-over	back-in-cludes-phi-Y_lag; Y_t	over	Lagged	beta=1	phi,beta=1	500	500	500	500	FE	RMSE, TWFE, ADL	grid. ing, tim- ing.				explicitly depends on included treatment lags.
jensen	calibrated two-period DiD by X,V; post controls Zm (mediator affect like) and (collider-like) and post shock U enter the two-period potential-outcome design.	D->Zm->Y; D->Zc->Y; U-base-line X,V	ATT	Default	n=5,000	500	500	500	500	FE	base-shifts, Bernoulli line, TVC, safe against pre-treatment, controlled, mod-els, full model					supplied CLI options mean actual execution should be established from saved scenario/result meta-data.

investitor	simulation	family	causal	estimator	description	Nr	Value	repetition	seed	effect	test	models	descript	file	paths	notes
mechanical	Galadriel	Dynamic	Z_{t-1} -> D_t and Z_{t-1} -> Y_t ; D_t -> Y_t ; D_t -> Z_{t-1} ;	direct beta=1	Grid doc-umented in source and tim-ing CSV; N_REPS=500.	500	500	iid	unit	8-bias, simulation	RMSE/MC	ADL model	ADL model	ADL model	ADL model	ADL model
nl_carry-over	Dynamic	Lagged	dy_{t-1} and dy_{t-2} ;	direct beta=1	N_REPS=500	500	500	iid	unit	FE. and RMSE, ADL MCSE, vari-ants.	FE. and RMSE, ADL MCSE, vari-ants.	FE. and RMSE, ADL MCSE, vari-ants.	FE. and RMSE, ADL MCSE, vari-ants.	FE. and RMSE, ADL MCSE, vari-ants.	FE. and RMSE, ADL MCSE, vari-ants.	FE. and RMSE, ADL MCSE, vari-ants.

investitor	title	family	causal_estimation	description	Nr_Tables	repetition	predecessors	effect	nodes	descript	file_path	status	notes
nl_colider	Dynamic	Lagged	beta=1	N_REPS=500	500	iid	unit/TWFE	Bias, simulation	ADL	RMSE, MCSE	simulations/ready/colider/	1	verified; numerical ranges preserved in versioned timing/results CSVs.
nl_colider	dual-role	Z-con-	T={10,80}	CSV		Gaus-sian	FE.	and	ADL	MCSE			
/	DGP	founder/current				in-		vari-					
fire-wall	with nonlin-	Z-col-	non-lin-			no-		ants	tim-				
	ear	lider	ear			va-		via	ing.				
	D->Z	un-	grid			tions;		shared					
	and/or	der	in			un-		es-					
	Y->Z	non-	tim-			sta-		ti-					
	func-	lin-	ing			ble		ma-					
	tions	ear	CSV.			draws		tor					
	(log2, log4,	chan-				han-		util-					
	soft-	nels.				dled		ity.					
	poly2,					in							
	power1.5,					source.							
	sin, in-												
	vlogit,												
	tanh,												
	Dlog,												
	Y2).												
nl_incolider	Dynamic	Lagged	beta=1	N_REPS=500	500	iid	unit/TWFE	Bias, simulation	ADL	RMSE, MCSE	simulations/ready/incolider/	1	short-hand; source is controling specification.
	D-	dual-	Z	grid	CSV	Gaus-	FE.	and	ADL	RMSE, MCSE			
	by-	role	con-	delta_DH		sian.							
	history	DGP	founder/current					vari-					
	in-	with	col-	rho_Z				ants.					
	ter-	inter-	lider,	in									
	ac-	action	non-	tim-									
	tion	term	lin-	ing									
		delta_DH		CSV.									
	(see	inter-											
	source)	ac-											
	in Z	tion.											
	pro-												
	cess.												

investitor	equation	family	causal	estimator	description	Nr	Value	repetition	spreads	effect	sectors	nodes	descript	file	paths	notes
over-control	D+rho	D	D	D	D	N=100,1000	1000	5000	deps	unit/	FW	F	bias	simulations	times	over-control
pure-	D_lag+u;	>Z_t-	beta=1	de-	Gaus-FE.	ADL	rel-									source
mediator	Z=alpha_Z	Yf(D)	linho_Z	fault	faultsian;		with	a-								de-
over-	Z_lag+nume-	ear	500;	(en-	pi-		D/Y	tive								clares
control	Y=alpha_Y	beta	to-	theta=.5,	delta_D=	14,	rho	D=rho	lags=.5	to						an en-
	D+theta	tor;	tal	non-	ron-	only	and	di-								viron-
	Z+rho_Yex-	ef-	lin-	mental-	con-	rect/linear										ment
	Y_lag+e.plic-	fect	ear	overi-	tem-	to-										over-
	itly	beta+theta		ride)	brates	po-	tal									ride;
	no	delta_D		scale,	ra-	tar-										execu-
	Y_t-	source.		not	ne-	gets;										tion
	>Z_t.			a	ous	FWL										meta-
				Monte	or	dif-										data
				Carlo	lagged	fer-										should
				re-	Z.	ence.										be
				sult.												con-
																sulted
																for
																actual
																reps.
over-control	D+rho	D	D	D	D	N=100,1000	1000	5000	deps	unit/	FW	F	bias	simulations	times	over-control
mediator	Z=alpha_Z	Yf(D)	linho_Z	fault	faultsian;		with	a-								de-
over-	Z_lag+rho	ZD	to-	2	Gaus-FE.	TW	F	MSE,								CSV is
control	D_lag+u;	>Y_t	tal=beta	theta	sian;	mod-	FWL									pre-
with	Z=alpha_Z	delta	D	D	pi-	els	shift,									served
non-	D+f_nl(D)	rho	Zi-	x	lot	with	grid/console									as a
lin-	Z_lag+nume-	tor;	rect=beta	NL	scale	con-	di-									first-
ear	Y=alpha_Y	beta	non-	states	cal-	tem-	ag-									class
D-	D+theta	tional	lin-	x	i-	po-	nos-									raw
to-	Z+gamma	Z_lag	ear	rho_Z	bra-	ra-	tics.									design
Z	Z_lag+rho	Y	to-	{.5,.7};	tion.	ne-										arti-
	Y_lag+e.found-	ing;	is	CSV.		ous										fact.
		no	model-			vs										
	Y_t-	specific.				lagged										
	>Z_t.					Z.										

investitor	family	causal	estimator	description	Nr	Value	repetition	fixed	effect	models	descript	file	paths	notes
persistent	unobserved	U_t	beta=1	N_REPS=500	500	iid	unit/standard	bias/MCSE	Estimations/dyns/diags/sims/	residuals/	residuals/	residuals/	residuals/	residuals/
con-	U_lag+v;	>D_t	ora-	36	CSV	Gaus-	FE.	9-	or-					known
tem-	D_t=...	kappa	cle	sce-		sian		modela-						limita-
po-	U_t+u;	U_t-	mod-	nar-		AR(1)		bat-	cle					tion:
ra-	Y_t=...	delta	els	ios		U		tery	con-					ADL
ne-	U_t+e;	con-	with			and		+	trast;					cannot
ous	Z_t=...	delta	Di-	kappa,phi_U,rho_Z,delta_U		or-		dis-						re-
un-	D_t+delta	rho_Y	tion	grids		other		a-	carded					move
ob-	Y_t+...	rho	on	doc-		in-		cle	draws/timing.					U_t
served		ous	U.	u-		no-		U						con-
con-	con-			mented		va-		mod-						found-
found-	found-			in		tions.		els						ing
ing	ing,			source.				+						with-
	along-							lag-						out
	side							ID						ob-
	dual-							sen-						serv-
	role							si-						ing U
	Z.							tiv-						or a
								ity						design.
								mod-						
								els.						
staggered	Diags	Staggered	delta=1	N_REPS=500	500	iid	unit/standard	bias, simulations/dyns/diags/sims/	residuals/	residuals/	residuals/	residuals/	residuals/	residuals/
stag-	ab-	bi-	con-	prob_n	CSV,rho_Y,delta_U	Gaus-	FE.	ZTWFE/MCSE,						the
gered	sorb-	nary	tem-	grid		sian		bat-	dis-					dynamic-
treat-	ing	D; D	po-	in		in-		tery.	carded					staggered
ment	after	his-	ra-	tim-		no-			draws/timing.					design
with	ran-	tory	ne-	ing		va-								rele-
post-	dom	af-	ous	CSV.		tions								vant to
treatment	drop-	fects	struc-			plus								lagged-
lag	tion	Y_lag,	tural			bi-								outcome
	for					nary								adjust-
	treated	dual-	effi-			treat-								ment.
	units;	role	cient,			ment								
	dy-	Z	not			as-								
	namic	may	nec-			sign-								
	Y,Z	be	es-			ment/adoption.								
	system	col-	sar-											
	in-	lider	ily a											
	cludes	on/off.	generic											
	lagged		event-											
	out-		study											
	come,		ATT.											
	so													
	Y_lag													
	may be													
	post-													
	treatment.													

inventory	family	causal	estimator	description	Nr	Value	repetition	fixed	effect	nodes	description	file	path	status	notes
v1_legacy	Multiple	Mixed;	varies	n=10,000	fixed	500	iid	varies	varies	means	/bias	simulations/v1_twfe	grid	no	considered
legacyex-ploratory	in-spect	by	in	early	static	Gaus-		com-							ver-
ploratory	cross-section,	source	sec-	sec-		u- where		par-							sioned
bun-	grid,	tions		tions;		ments	down.	isons							CSV is
dle	ADL,	be-		later		blocks		printed							writ-
	rho,	fore		N=200,T=20,				in-							ten by
	and	reuse.		nsim=500;				session.							this
	common-			beta											script;
	warm-			val-											re-
	up			ues											tained
	DGPs			1 or											in
	in one			5.											inven-
	script;														tory to
	no														pre-
	single														vent
	per-														unsup-
	sisted														ported
	output														quan-
	con-														tita-
	tract.														tive
															claims.
v1_twfe	Y=beta	D-	beta	400	source	500	iid	unit	TWFE	Bias, simulation	simulations/v1_twfe	grid	no	considered	Re-
base-	D+unit	>Y;	and	sce-	CSV	CSV	stat	and	short	IVB/FWL					CSV is
line	FE+time	D-	FWL	nar-			sce-	sian	time	Y~D; iden-					the
FWL/	FWFEs;	>Z<-	spec-	ios x			nar-	ion-	FE	long tity,					au-
grid	Z=(gamma	Y_D+gamma	gamma_500	no-			no-	in	Y~D+Z	FE					thori-
	beta)D+unit	time	ca-	reps;			va-	short	aux-	and					tative
	com-	tem-	tion	gamma_Y,			tions;	long,	il-	sta-					pa-
	po-	pora-	shift	gamma_D,			gen-	and	iary	bil-					rame-
	nents+nu	ne-	beta_log	gamma,			er-	aux-	Z~D.	ity					ter
	(as	ous	beta_within	short-			ated	il-	fields						grid;
	imple-	col-		R2			unit/time	any	in						legacy
	mented).	lider.		grid			com-	mod-	CSV.						design
				(see			po-	els.							de-
				re-			nents.								scribed
				sult											as
				CSV).											tauto-
															logical
															in
															reposi-
															tory
															in-
															struc-
															tions.

[illegible]

5 Static validation record

The builder checks the required inventory fields, unique identifiers, existence of every source script and declared result artifact, and records artifact size, header, and MD5. For numeric parameter columns present in artifacts, it checks probabilities/persistence terms on $[0,1]$ and positive count fields; for date-like columns it rejects unparsable dates and dates after the audit date. It does not execute simulation functions, therefore it makes no new Monte Carlo draws. The following gaps remain material:

1. `sim_ivb_completa.R` contains legacy exploratory simulations without a versioned persisted result file, so it cannot support a numerical manuscript claim.
2. Some scripts allow repetitions to be overridden by environment variables or command-line arguments. The source default is reported; actual execution settings should be recovered from the saved result/timing artifact before quoting a number.
3. The inventory maps source scripts to artifacts by their explicit output paths. It does not establish Git commit ancestry for historical result files; MD5 values make the audited artifact state reproducible.